



# PULSE

Profit Pool Intelligence Engine

A Bayesian simulation platform that transforms  
strategic force assessment into decision intelligence.

# Where will your **profit pools** be in five years?

The companies that answer this question first  
will define the next decade of consumer goods.

# Profit pools are shifting — fast

## Regulation

EU Green Deal, PFAS bans, EPR mandates are rewriting category economics

## Technology

AI, biotech, DTC platforms disrupt every step of the value chain

## Sustainability

Climate costs, water scarcity, packaging levies hit the bottom line

## Consumer

Clean beauty, premiumization, value polarization reshape demand curves

## Retail Power

Private label growth and channel consolidation squeeze branded margins

## Competition

Challenger brands, cross-category entry, and M&A redraw the competitive map

# Most companies react. Leaders anticipate.

## Typical Approach

- Static, annual assessments
- Point estimates without uncertainty
- Forces analyzed in isolation
- No competitive response modeling
- Backward-looking calibration gap

## What's Needed

- ✓ Living, probabilistic simulation
- ✓ Full uncertainty quantification
- ✓ Causal force propagation
- ✓ Game-theoretic competitor modeling
- ✓ Backtested for credibility

# PULSE

A privacy-first, AI-augmented simulation engine that models how strategic forces reshape profit pools across categories and time.

## 01 Simulate

Bayesian Monte Carlo  
with 10,000+ scenarios

## 02 Connect

Causal DAG maps how  
forces propagate

## 03 Decide

Resource allocation  
optimizer for action

# How PULSE works



# Six forces shape every profit pool

## Consumer

Demand shifts, premiumization, clean beauty, health consciousness

## Customer

Retail consolidation, private label growth, channel power dynamics

## Technology

AI/biotech, DTC platforms, smart manufacturing, digital commerce

## Government

EU Green Deal, PFAS regulation, EPR mandates, compliance costs

## Environmental

Climate impact, water scarcity, packaging levies, sustainability pressure

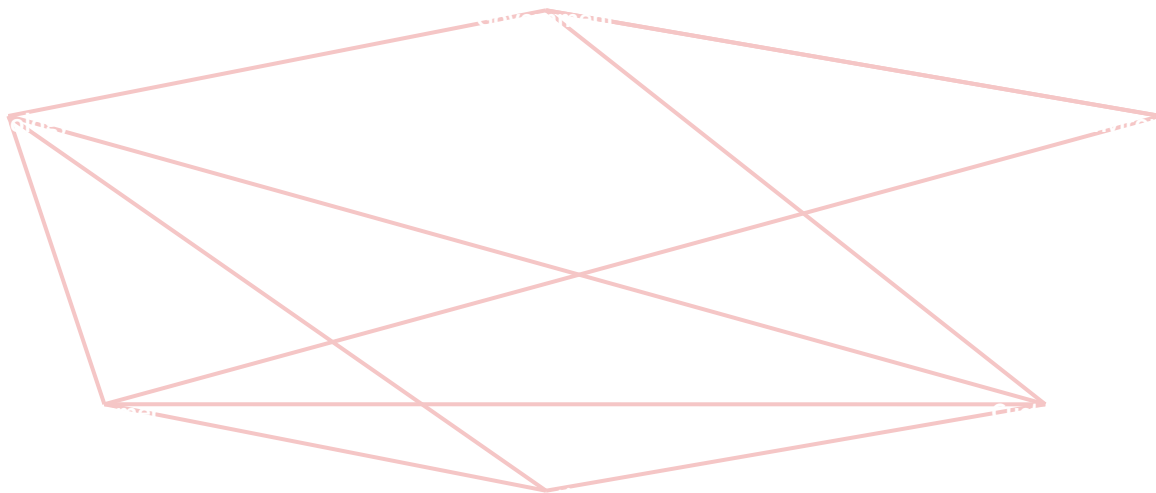
## Competitive

Challenger brands, M&A activity, cross-category entry, price wars

*Each force contains 8–12 individually scored trends — 60+ in total across all categories.*

# Forces don't act alone — they propagate

PULSE uses a Causal DAG to model how a shock in one force propagates to others — with calibrated weights and time lags.



Example: EU regulation (Gov) → reformulation costs (Tech) → shelf price (Customer) → demand shift (Consumer)

# Not guesses — Bayesian probabilities

Traditional approaches give you a single number.

PULSE gives you a full probability distribution.

## Bayesian Priors

Start with expert knowledge, update with evidence

## Copula Dependencies

Model how risks cluster together in crises

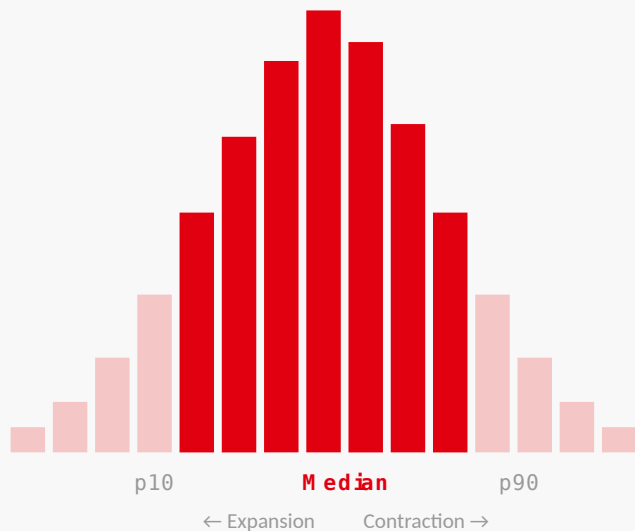
## 10,000+ Simulations

Full posterior distribution, not a point estimate

## Convergence Tested

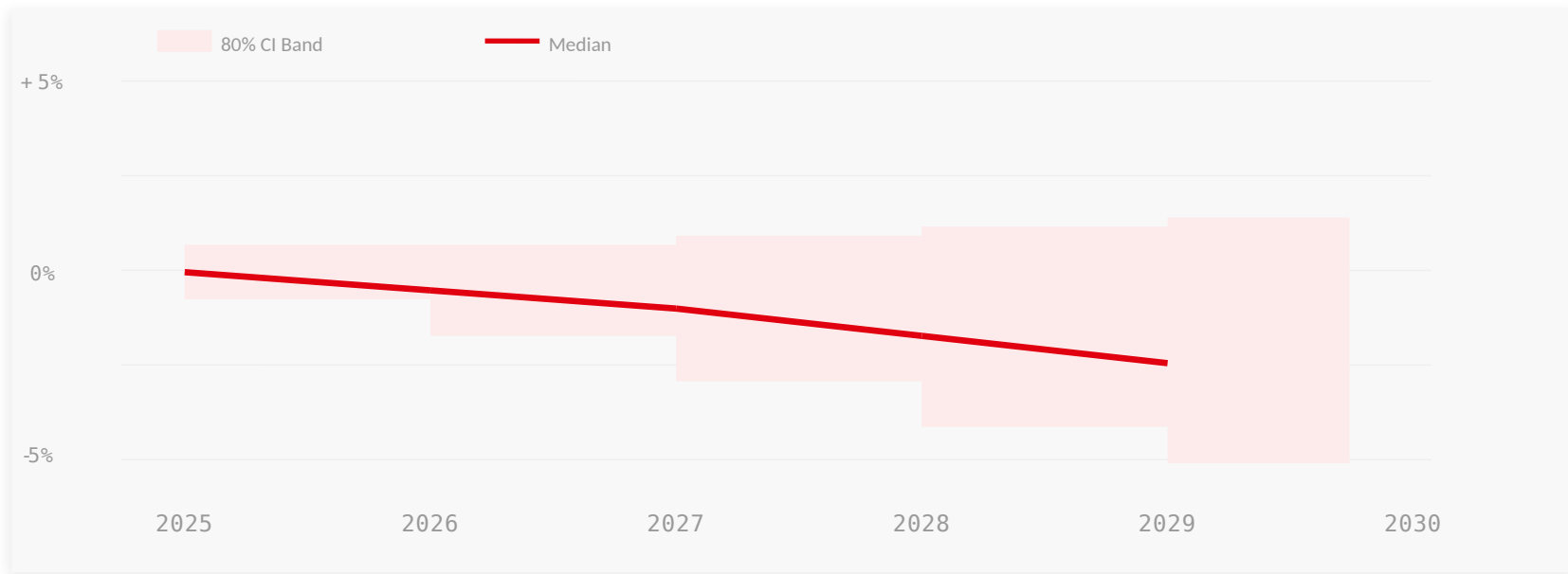
Gelman-Rubin  $R$ -hat < 1.05 for statistical rigor

Posterior Distribution



# Five-year paths, not two-point snapshots

PULSE models continuous trajectories from 2026 to 2030. Velocity tracking and early-warning triggers enable path-dependent strategy.



# One output: the Shift Matrix

A single table of percentage shifts by category, year, and confidence level. Apply directly to financial models.

| Category    | 2026  | 2027  | 2028  | 2029  | 2030  |
|-------------|-------|-------|-------|-------|-------|
| Hair: Color | -0.3% | -0.8% | -1.4% | -2.3% | -3.2% |
| Hair: Care  | +0.2% | +0.5% | +1.1% | +1.8% | +2.6% |
| LHC: FCN    | -0.5% | -1.2% | -2.1% | -3.5% | -4.8% |
| LHC: ADW    | +0.1% | +0.4% | +0.9% | +1.5% | +2.1% |
| LHC: IC     | +0.0% | +0.1% | +0.2% | +0.3% | +0.4% |

Values shown are median shifts. Full output includes p10, p25, p75, p90 bands plus causal decomposition.

Application:  $GP1_{\text{projected}} = GP1_{\text{actual}} \times (1 + \text{shift} \%)$

# The Delphi Process: structured expert scoring

Replacing ad-hoc scoring with a rigorous, debiased, multi-round elicitation protocol.

Pre-Workshop

## Calibration

Present known-outcome historical trends to each scorer. Detect systematic biases (optimism, anchoring, recency). Compute individual calibration factors.

Round 1

## Independent Scoring

Each expert scores impact (1-5) and probability (1-5) for every trend — blind, without seeing others' scores. Documented rationale required.

Round 2

## Calibrated Re-Scoring

Share anonymized score distribution across all scorers. Experts re-score with group context. Measure inter-rater reliability (Krippendorff's  $\alpha$ ).

Round 3

## Consensus & Debiasing

Discuss outlier scores in structured session. Apply bias correction factors. Final score = weighted median by calibration accuracy.

Quality Gate:  $\alpha > 0.67$  required | Bias-corrected | Full audit trail | Scores feed Bayesian priors

# Competitors will respond — we model that

A game-theoretic layer models likely competitive responses. Iterative Nash equilibrium converges to a stable outcome.

## Premium Defender

- Innovation-led defense
- Premiumize further
- Lobby & comply early

## Sustainability Leader

- Portfolio rationalization
- Lead compliance
- Acquire disruptors

## Category Specialist

- Defend core categories
- Selective tech adoption
- Minimum compliance

*All inputs are public market intelligence — no competitor financials enter the model.*

# From analysis to action: resource allocation

*Every output answers: so what should we do?*

| INVEST MORE                                                                | DEFEND                                                                                 | HARVEST                                               |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Hair: Care</b><br><b>LHC: ADW</b><br><br>Growing pools<br>with momentum | <b>LHC: FCN</b><br><b>Hair: Color</b><br><br>Contracting pools<br>requiring protection | <b>LHC: IC</b><br><br>Stable pools<br>with low growth |

*All recommendations are in relative weights — never absolute currency amounts.*

# The model earns credibility before it predicts

PULSE uses historical versions (V1–V11) and actual market data to calibrate parameters. It proves accuracy before predicting.

- Attenuation factor (derived, not assumed)
- Distribution shapes (learned from prediction errors)
- Causal DAG weights (fitted to historical co-movements)
- Correlation structure (empirical, not flat)
- Overall accuracy score (reported to ExCo)

## Backtesting Report

**73%**

of historical shifts predicted  
within the 80% confidence interval

Based on V1–V11 historical versions  
vs. actual Euromonitor market data

# Security by architecture, not by policy

## ENTERS PULSE

- ✓ Trend names & descriptions
- ✓ Impact / probability scores (1-5)
- ✓ Direction (expansion / contraction)
- ✓ Category names (public knowledge)
- ✓ Causal propagation weights
- ✓ Public market data (Euromonitor)

## NEVER ENTERS PULSE

- ✗ Net External Sales (NES)
- ✗ Gross Profit (GP1, GP2)
- ✗ Regional financial splits
- ✗ Cost structures (COGS, OpEx)
- ✗ Company market shares
- ✗ Management unit financials

*Financials are structurally excluded — enforced by schema, firewall, and automated tests.*

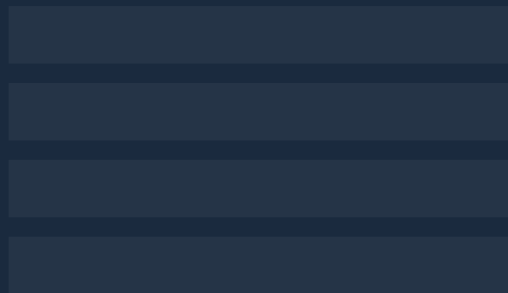
# The War Room: one screen tells the story

NetPoolShift: -3.1% | 80% CI: [-1.5% , -5.5% ] | Accuracy: 73% | Velocity: Accelerating

Force × Category Heatmap



Contextual Panel



Path Timeline 2026 — 2030



# Scenario engine: shock, propagate, decide

|                           |                                                                       |
|---------------------------|-----------------------------------------------------------------------|
| <b>Base Case</b>          | Current expert scores, causal DAG active, no external shocks          |
| <b>Green Squeeze</b>      | Environmental shock → Government, Technology, Customer, Consumer      |
| <b>Tech Disruption</b>    | Technology shock → Consumer adoption, Competitive gaps, Channel shift |
| <b>Price War</b>          | Competitive shock → Customer margin pressure, Consumer trading down   |
| <b>Regulatory Cascade</b> | Government shock → propagates through DAG with lag structure          |
| <b>Perfect Storm</b>      | Correlated tail event — t-copula at 1st percentile (crisis scenario)  |

*Custom scenarios: user shocks a specific force, the Causal DAG propagates the rest automatically.*

# AI augments judgment — never replaces it

## Trend Scanner

Monitors 19 free-tier data sources continuously.  
Flags new developments. Weekly digest.

GDELT · GNews · RSS · ECHA · EPO · Google Trends

## Score Calibrator

Cross-validates expert scores for consistency.  
Benchmarks against real-world trajectory data.

Internal consistency + external benchmarks

## Scenario Narrator

Converts Shift Matrix into executive narrative.  
Explains causal propagation in plain language.

LLM-generated from relative data only

## Chat Interface

Natural language queries about the model.  
Financial queries get redirected.

Conversational AI layer

*Provider-agnostic: Claude API (MVP) → Azure OpenAI (production) via single config change.*

# Built on proven, enterprise-grade technology

## Simulation Engine

Python 3.11+ , NumPy , SciPy , Pandas , NetworkX

Bayesian Monte Carlo, copula dependencies, causal propagation

## API & Data

FastAPI, Pydantic , SQLAlchemy , OpenPyXL

REST API, validation, zero-financial schema, audit trail

## War Room UI

React 18 , Vite , D3 , Recharts , Tailwind , Framer

Single-page War Room, 60fps animations, Apple-grade UX

## AI Layer

LLM Provider ABC , Claude / Azure OpenAI

Trend scanning, score calibration, executive narration

**Total running cost: €0/year — All 19 external APIs are free tier. No vendor lock-in.**

# Building with DX: from MVP to enterprise

## MVP

Strategy Team

Working prototype:  
Bayesian engine, CLI,  
Shift Matrix output.  
Proves the concept.

## DX BUILD

Joint Team

Cloud deployment,  
War Room dashboard,  
API integration,  
Security hardening.

## ENTERPRISE

DX Owned

Azure tenant hosting,  
SSO + access control,  
Automated pipelines,  
ExCo reporting.

## DX Engagement Steps

- 1 Handoff: MVP code + spec to DX
- 2 Azure tenant setup + CI/CD pipeline
- 3 Security review + InfoSec approval
- 4 Dashboard deployment + SSO
- 5 Financial system integration
- 6 Ongoing: quarterly model updates

# Phased delivery — standalone value at each step

1

2

3

## Historical Calibration

Embedded

Backtest V1-V11,  
calibrate parameters,  
earn credibility

## Bayesian Engine (CLI)

Months 1-3

Monte Carlo simulation,  
causal DAG, optimizer,  
Shift Matrix output

## War Room Dashboard

Months 4-6

React UI, heatmap,  
path visualization,  
export center

## AI Intelligence Layer

Months 7-9

Trend scanner, calibrator,  
narrator, chat interface,  
automated reporting

Each phase delivers standalone value. Phase 1 alone provides a probabilistic, causally-structured simulation. Every subsequent phase compounds the advantage.

# The Ask

- 1 Endorse PULSE as the strategic profit pool simulation platform
- 2 Assign DX resources for cloud deployment (Phase 2-3)
- 3 Sponsor a pilot run with one business unit (Hair or LHC)

The MVP is built. The methodology is proven.

The question is not whether to simulate — it's how fast we can scale.